

Homework Modern Physics

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For exercises from the textbook, the corresponding number is reported. The text may have been edited for clarity.

Some of the following exercises (possibly with different numbers) may be on the **mini-test** taking place on the Monday of the following week, or on the **midterms**.

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WEEK II

STRUCTURE OF THE ATOM & WAVE PROPERTIES OF MATTER AND QUANTUM MECHANICS I

1. Shortest wavelength emitted by light atoms / Example 4.9

Calculate the shortest wavelength that can be emitted by the H atom. Compare it with that of the Li^{++} ion.

Hint: the shortest wavelength to which pair (n_u, n_ℓ) corresponds?

2. Adsorption transition / Ex. 4.30

Light from a Nd:YAG (Neodymium-doped Yttrium Aluminum Garnet) laser with a wavelength of 397 nm is incident upon a hydrogen atom in the $n = 2$ state at rest. What is the highest state to which hydrogen can be excited?

3. Spectrum of positronium / Ex. 4.56

(a) Calculate the energies of the four lowest states of positronium. (b) Determine the wavelengths of the K_α , K_β , L_α , and L_β transitions (cfr. Ex. 4.32 and its solution in this week's Tutorial).

Notation: Historically, the stationary states were also given letter identifications: K shell ($n = 1$), L shell ($n = 2$), M shell ($n = 3$), and so on. The x-rays emitted when an atom changes energy states are given different names depending on the initial and final states. The Greek letter subscripts indicate the value of $\Delta n = n_u - n_\ell$, and the roman letters the value of n for the final state. Examples: K_α denotes the emission transition between the $n = 2$ and $n = 1$ states; L_β between $n = 4$ and $n = 2$; (cfr. Fig. 4.18 in the textbook).

4. De Broglie wavelength / Ex. 5.13

Find the kinetic energy of (a) photons, (b) electrons, (c) neutrons, and (d) α particles that have a de Broglie wavelength $\lambda = 0.13$ nm.

Hint: for the photon, kinetic energy equals total energy, for the other particles, assume the non-relativistic limit.

5. Probing the nuclear structure

What is the minimum energy of a photon that allows probing the internal structure of the nucleus of iron? Assume the nucleus it is a sphere of radius $r_{\text{Fe}} \approx 4.0$ fm.